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SECTION A

- Find the value of k for which the roots of the quadratic equation $(k-5)x^2 + 2(k-5)x + 2 = 0$ are equal.
- Find the value of y for which the distance between the points $(2, -3)$ and $(10, y)$ is 10 units.
- Write whether the rational number $13/3125$ has a decimal expansion which is terminating or non-terminating repeating.
- Write the n th term of the A.P. $1/k, (1+k)/k, (1+2k)/k$.
- If $\sin x + \cos x = 2 \cos 90$, find the value of x .
- DE is drawn parallel to the base BC of triangle ABC , meeting AB at D and AC at E . If $AB/CD = 4$ and $CE = 2$ cm, find AE .

SECTION B

- A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is three times that of drawing a red ball, find the number of blue balls.
- The 5th and 15th terms of an A.P. are 13 and 17. Find the sum of the first 21 terms.
- Using Euclid's Division Algorithm, find the HCF of 225 and 867.

- If the point $(0, 2)$ is equidistant from the points $(3, k)$ and $(k, 5)$, find k .
- Find the value of a for which the pair of linear equations $2x + 3y = 7$ and $4x + ay = 14$ has infinitely many solutions.

SECTION C

- Show that any positive odd integer is of the form $4q + 1$ or $4q + 3$.
- The tens digit of a number is twice its units digit. The number obtained by interchanging the digits is 36 less than the original number. Find the original number.
- Show that 1, $1/2$ and 2 are the zeroes of the polynomial $2x^3 + x^2 - 5x + 2$.
- Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre.
- S and T are points on the sides PR and QR of triangle PQR such that $\angle P = \angle RTS$. Show that triangle RPQ is similar to triangle RTS .